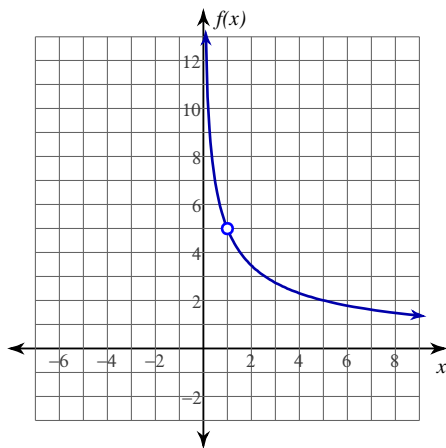


L'Hôpital's Rule

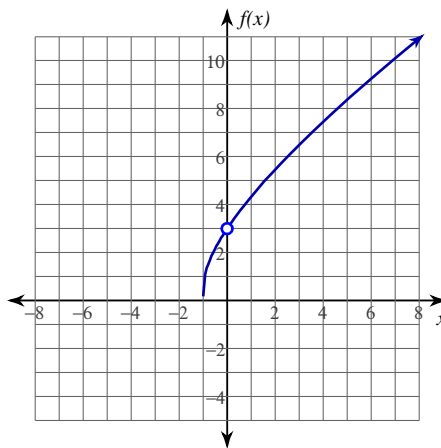
Date _____ Period _____

Evaluate each limit using L'Hôpital's Rule.

1) $\lim_{x \rightarrow 1} \frac{5 \ln x}{x - 1}$



2) $\lim_{x \rightarrow 0} \frac{3x}{\ln(x + 1)}$



3) $\lim_{x \rightarrow 0^+} 5x^2 \ln x$

4) $\lim_{x \rightarrow \infty} 4x \cdot e^{-x}$

5) $\lim_{x \rightarrow \frac{\pi}{2}} (3 \sec x - 3 \tan x)$

6) $\lim_{x \rightarrow \infty} \left(\frac{x^2}{x - 1} - \frac{x^2}{x + 1} \right)$

7) $\lim_{x \rightarrow 0^+} 5 \cdot (\tan x)^{\sin x}$

8) $\lim_{x \rightarrow 0^+} 3x^x$

Evaluate each limit. Use L'Hôpital's Rule if it can be applied. If it cannot be applied, evaluate using another method and write a * next to your answer.

9) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x}$

10) $\lim_{x \rightarrow 0^+} \frac{e^x + e^{-x}}{\sin(2x)}$